ClubSync: 15-Minute Multi-Speaker Presentation Script

Format: 15-Minute Word-for-Word Live Script (900s) + 5-Minute Evaluator Viva
Order: Member 1 ➔ Member 2 ➔ Member 3 ➔ Member 4 ➔ Member 5 ➔ Member 6 ➔ Member 1 (Wrap-Up)

Segment 1: Member 1 (Team Leader — Architecture & Product Design)

Time Allocation: 00:00 - 02:30 (2 mins 30 secs)
Visual / Screen Cue: *Presentation Title Slide ➔ Live System Architecture Diagram ➔ Database Schema ERD*

"Respected panel, professors, and fellow colleagues. Good morning/afternoon.

I am **[Leader Name]**, leader of Team ClubSync. Today, our team is proud to present **ClubSync** — the Unified Collegiate Society and Event Operations Platform, architected specifically for high-density engineering environments like PICT.

Every semester, campus societies like IEEE, ACM, CSI, and Robocon host flagship hackathons, symposia, and workshops. Yet, behind the scenes, student officers face operational chaos: registrations rely on fragile Google Forms; payments bounce between personal UPI QR codes with zero financial auditability; attendance desks struggle with chaotic physical queues; and club governance happens informally over WhatsApp.

As the lead architect and product designer, my objective was to engineer a robust, enterprise-grade web ecosystem capable of eliminating these pain points entirely.

Our architecture combines a reactive **Vite and React 18** client with a hardened **Node.js 20 Express** REST engine, backed by **Neon Serverless PostgreSQL**. On the database tier, we designed a normalized relational schema of 18 tables with strict foreign key constraints and ACID guarantees. Crucially, we enforce a strict 100% parameterized SQL query model with zero string interpolation, neutralizing SQL injection vectors.

On the security perimeter, our endpoints are protected by Helmet security headers, rate limiters, bcrypt-hashed credentials, and custom Broken Object-Level Authorization (BOLA) middleware that strictly fences sensitive student data and financial ledgers to authorized club officers.

I will now hand over to **[Member 2]** to demonstrate our revolutionary Event Day Kiosk and gate check-in pipeline."

Segment 2: Member 2 (Event Operations & Hardware Integration)

Time Allocation: 02:30 - 05:00 (2 mins 30 secs)
Visual / Screen Cue: *Live Demo on localhost:5173 ➔ Log in as officer@clubsync.app ➔ Event Detail ➔ 'Desk Kiosk Mode' ➔ Scan Barcode*

"Thank you, [Leader Name]. Good morning everyone. I am **[Member 2]**, and I led the Event Day Operations and Hardware Integration modules.

On event morning at any major college auditorium, registration desks become bottlenecks. Volunteer laptops struggle with slow network forms, leading to 45-minute queues.

To solve this, I built the **ClubSync Event Day Desk Kiosk** [pointing to screen]. This is a high-throughput, hands-free operational terminal.

Notice three key technical innovations:
1. First, the **Continuous Scanner Wedge Listener**: Built directly into the DOM lifecycle, it automatically recaptures focus on any incoming USB or Bluetooth barcode/QR scanner stream in milliseconds. As an attendee presents their digital pass, the scanner triggers an instant lookup, plays an audio synthesizer chime, and logs check-in in under 400 milliseconds.
2. Second, **Instant Thermal Badge Printing**: Integrated using CSS @media print rules calibrated specifically for 80-millimeter thermal receipt printers. With one tap or upon auto-scan, it prints a physical delegate badge complete with student name, PRN, branch, team name, and a scannable verification barcode.
3. Third, **Spot Walk-In Registration**: For students who show up unregistered at the auditorium doors, officers can register them in less than 15 seconds through our walk-in pipeline, which simultaneously registers the record, generates their credential ID, and prints their badge.

Next, **[Member 3]** will showcase how we overhauled financial compliance and corporate sponsorships."

Segment 3: Member 3 (Financial Systems & Sponsorship CRM)

Time Allocation: 05:00 - 07:30 (2 mins 30 secs)
Visual / Screen Cue: *Click 'Register / Pay ■100' ➔ Open YouPI Payment Modal ➔ Club Detail ➔ Sponsorships CRM Kanban Board*

"Thank you, [Member 2]. I am **[Member 3]**, responsible for ClubSync's Financial Systems and Sponsorship CRM.

Commercial payment gateways charge 2 to 3 percent per transaction and require complex institutional GST registrations, which student clubs do not possess. Most clubs resort to unofficial student Google Pay numbers, creating major accounting discrepancies during annual Gymkhana audits.

To eliminate gateway fees entirely, I developed the **YouPI Zero-Fee Gateway** [pointing to modal]. It dynamically generates deep-linked UPI intents for GPay, PhonePe, and Paytm pointing directly to the club's designated account. When a student pays, they input their 12-digit Bank UTR reference. The transaction is logged into a tamper-evident audit ledger, and the officer verifies it with one click, issuing an official collegiate e-receipt.

Next, moving beyond payments, collegiate symposia survive on corporate sponsorships. Traditionally, brand outreach is tracked across fragmented personal spreadsheets.

We implemented the **Corporate Sponsorship & Brand Outreach CRM** [pointing to Kanban board]. Built as a 5-stage Kanban pipeline—from initial Lead to Funds Received—it tracks active deal values, signed MOUs, and tier milestones. When an officer selects a sponsorship tier—such as Title or Associate Sponsor—our engine automatically activates the required contractual deliverables: auditorium standee placements, logo on banners, keynote speaking slots, and resume disclosures.

I now invite **[Member 4]** to present our Democratic Governance and Mentorship framework."

Segment 4: Member 4 (Governance, Mentorship & Academic Compliance)

Time Allocation: 07:30 - 10:00 (2 mins 30 secs)
Visual / Screen Cue: *Switch to 'By-Laws & Resolutions' Tab ➔ Side-by-side red/green diff ➔ Mentorship Tab ➔ OD Letter Modal*

"Thank you, [Member 3]. Good day everyone, I am **[Member 4]**, and I engineered ClubSync's Governance, Peer Mentorship, and Academic Compliance infrastructure.

Student societies are democratic institutions, yet their constitutions often sit as static, forgotten PDF files, with amendments passed informally.

In ClubSync, we pioneered the **Constitutional By-Law Amendment Portal** [pointing to screen]. Officers can formally table amendments to club constitutions. The interface displays an automated side-by-side red and green diff comparing the existing constitutional clause with the proposed revision.

More importantly, our backend enforces mathematical democratic thresholds: amendments require a strict 66.6% supermajority and valid quorum calculation before they can be ratified into club law. Every vote is recorded with mandatory rationale comments for institutional transparency.

Next is our **Senior-Junior Mentorship Pairing Hub**: Senior students in Third and Final Year publish mentor profiles tagged with technical competencies like DSA, Web3, and Robotics. Junior students can book 1-on-1 pairing slots with automatic campus venue or Google Meet routing, and leave verified star ratings that reward mentors with AICTE activity hours.

Finally, addressing academic compliance: we built the **Automated On-Duty (OD/DL) Leave Letter Generator**. Rather than students manually writing applications, ClubSync auto-generates formal leave sanction letters for attendees, pre-formatted for department heads and faculty advisors with unique verification references.

I will now pass the stage to **[Member 5]** to present our Webhook syndication and AI systems."

Segment 5: Member 5 (Communications, Real-Time Systems & AI)

Time Allocation: 10:00 - 12:30 (2 mins 30 secs)
Visual / Screen Cue: *Open Club 'Webhooks & Syndication' Tab ➔ Send Test Ping ➔ Live Q&A; Projector View ➔ Club AI Assistant*

"Thank you, [Member 4]. I am **[Member 5]**, and I led our Communications, Real-Time Interaction, and AI subsystems.

Students rarely check emails for club updates; their attention lives on Discord and Telegram. Previously, publicity teams manually copied and pasted announcements across dozens of groups.

We built the **Multi-Channel Webhook Syndicate** [pointing to Webhooks tab]. The moment an officer publishes an event or urgent room change in ClubSync, our backend dispatcher translates the event payload into rich, color-coded Discord Embeds and formatted Telegram HTML posts, broadcasting them across connected channels simultaneously. We also included an emergency blast broadcaster with full delivery health logging.

Next, for symposium keynote sessions, we built the **Slido-Style Live Stage Q&A;** [pointing to Q&A; modal]. In an auditorium, passing around microphones is inefficient. With our live projector view, audience members submit and upvote questions in real-time. Speakers can pin highlighted questions and mark them answered live on stage.

Lastly, we integrated **Groq Llama 3 AI** into our backend. Operating over club meeting minutes and constitutional documents, it provides an intelligent conversational assistant that lets members ask natural language questions about club policies, past decisions, and event guidelines in sub-second response times.

I now invite **[Member 6]** to cover our Frontend Design System, PWA offline capabilities, and Public Verification Registry."

Segment 6: Member 6 (Frontend UI/UX, PWA & Public Credential Registry)

Time Allocation: 12:30 - 14:15 (1 min 45 secs)
Visual / Screen Cue: *Toggle Mobile View ➔ Show PWA offline behavior ➔ Navigate to /verify/PICT-CERT-2026-X89 ➔ Click 'Add to LinkedIn'*

"Thank you, [Member 5]. I am **[Member 6]**, and I focused on our Frontend Architecture, Progressive Web App engineering, and the Public Verification Registry.

From day one, our design philosophy was mobile-first, tactile, and zero-latency. Using **Tailwind CSS** and **Framer Motion**, we crafted an intuitive, dark-mode native interface with micro-interactions, responsive sidebars, and fluid drawer states.

Campus Wi-Fi in college auditoriums and basements is notoriously unreliable. To guarantee zero operational interruption, we configured ClubSync as a full **Progressive Web App (PWA)** using modern service worker precaching. Essential UI shells and cached registration tables remain interactive even when connection drops.

A major innovation is our **Public Certificate Verification Registry** [pointing to /verify/:id]. Certificate forgery is a widespread problem on campus resumes. In ClubSync, every event certificate carries a cryptographic ID and a QR code pointing to a public verification endpoint. Anyone—recruiters or faculty—can verify the attendee's name, PRN, event date, and digital institutional seal.

With our 1-click **'Add to LinkedIn'** integration, students can immediately export and embed their verified ClubSync credentials onto their professional LinkedIn profiles with pre-filled certification metadata.

I now hand back to our team lead, **[Member 1]**, for our project summary and closing remarks."

Segment 7: Member 1 (Team Leader — Project Summary & Closing)

Time Allocation: 14:15 - 15:00 (45 secs)
Visual / Screen Cue: *GitHub Repository / Project Metrics Summary Slide ➔ Thank You & Q&A; Slide*

"Thank you, [Member 6].

To summarize: in ClubSync, our team has built a production-ready, security-audited, full-stack campus platform spanning 20+ fully realized modules:
• Desk Kiosk check-in with continuous barcode scanning & thermal badge printing,
• Zero-fee UPI payments and corporate sponsorship CRMs,
• Democratic constitutional governance and peer mentorship pairing,
• Multi-channel Discord/Telegram webhooks and AI summarization,
• All backed by a secure PostgreSQL database with zero dependency vulnerabilities.

ClubSync modernizes the collegiate student society experience, setting a new standard for campus technology operations.

Thank you for your time. The floor is now open, and our team would be pleased to answer your questions."